

Chapter 7 — Physical limits

Tokens were the unit of meaning. Here they become the unit of price.

This is a **governing-constraint analysis**. You already do this with load paths and limit states: find the binding constraint, then reason about what relaxes it.

Symbols

All quantities positive definite — no sign convention is required. SI throughout.

Symbol	Meaning	Unit
B	batch size, tokens per forward pass	—
N_{act}	active parameters per token	—
N_{tot}	total parameters	—
F	arithmetic throughput	FLOP s^{-1}
M	throughput in multiply-accumulates, $1 \text{ MAC} = 2 \text{ FLOP}$	MAC s^{-1}
b_w	bytes per stored parameter	byte

The roofline bound

$$T \geq \max(T_{\text{comp}}, T_{\text{mem}})$$

$$T_{\text{comp}} = \frac{2 B N_{\text{act}}}{F} = \frac{B N_{\text{act}}}{M}$$

$$T_{\text{mem}} = \frac{\overbrace{N_{\text{tot}} b_w}^{\text{weight fetch}} + \overbrace{B L b_{\text{kv}}}^{\text{KV fetch}}}{\beta}$$

$$c \equiv \frac{T}{B}$$

$$\frac{T_{\text{mem}}}{B} = \underbrace{\frac{N_{\text{tot}} b_w}{\beta B}}_{\propto 1/B} + \underbrace{\frac{L b_{\text{kv}}}{\beta}}_{\text{constant in } B}$$

Dimensions. $2BN_{\text{act}}$ is a FLOP count, so T_{comp} is $\text{FLOP}/(\text{FLOP s}^{-1}) = \text{s}$. $N_{\text{tot}}b_w + BLb_{\text{kv}}$ is bytes, so T_{mem} is $\text{byte}/(\text{byte s}^{-1}) = \text{s}$. Cost c is s token^{-1} .

What the hyperbola says

The weight-fetch term carries $1/B$. At $B = 1$ every parameter in the model is read from memory to produce one token.

As B grows that term amortises and cost flattens onto the KV floor $L b_{\text{kv}}/\beta$, which B does not help — it is linear in context length and paid per token regardless.

Consequence. Long context is bounded by memory bandwidth and capacity, not by compute. That is a different claim from anything about accuracy, which has no equation and is Chapter 5's material.

Critical batch size

$$T_{\text{comp}} = T_{\text{mem}}^{(\text{weight})}$$

$$\frac{2 B^* N_{\text{act}}}{F} = \frac{N_{\text{tot}} b_w}{\beta}$$

$$B^* = \underbrace{\frac{F b_w}{2 \beta}}_{\text{hardware ratio}} \times \underbrace{\frac{N_{\text{tot}}}{N_{\text{act}}}}_{\text{sparsity}}$$

The source reports the hardware ratio as ≈ 300 , dimensionless and stable across several GPU generations. Critical batch size is then $\approx 300 \times \text{sparsity}$.

The denominator is the whole argument

T_{comp} was written twice above, and the two forms are not interchangeable:

$$T_{\text{comp}} = \frac{B N_{\text{act}}}{M} \quad M \text{ in MAC s}^{-1}$$

$$T_{\text{comp}} = \frac{2 B N_{\text{act}}}{F} \quad F \text{ in FLOP s}^{-1}$$

Substituting a spec-sheet FLOP s^{-1} into the MAC-denominated form makes the answer **wrong by exactly a factor of two**. An outside reader caught this in the transcript's comment thread.

A unit-definition error, in an expert lecture, found by peer review, producing a clean factor of two. Chapter 4 uses this as its anchor.

Trading compute for memory

Invertible architectures recompute activations instead of storing them.